MIT AI Risk Repository

FutureTech
THE ECONOMIC AND TECHNICAL FOUNDATIONS OF PROGRESS IN COMPUTING

Massachusetts Institute of Technology

AI Risk Taxonomies

MIT AI Risk Repository

OVERVIEW

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MIT AI Risk Repository - Causal Taxonomy of AI risks

Domain	Level	Description of how the risk is presented in evidence
Entity	AI	Due to a decision or action made by an AI system
	Human	Due to a decision or action made by humans
	Other	Due to some other reason or ambiguous
Intent	Intentional	Due to an expected outcome from pursuing a goal
	Unintentional	Due to an unexpected outcome from pursuing a goal
	Other	Without clearly specifying the intentionality
Timing	Pre-deployment	Before the AI is deployed
	Post-deployment	After the AI model has been trained and deployed
	Other	Without a clearly specified time of occurrence

"Malicious utilization of AI has the potential to endanger digital security, physical security, and political security. International law enforcement entities grapple with a variety of risks linked to the Malevolent Utilization of AI." Habbal, 2024 [29.03.01]



Entity = Human
Intent = Intentional
Timing = Post-deployment

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MIT AI Risk Repository - Domain Taxonomy of AI risks

Domain / Subdomain	
1	***Discrimination & Toxicity***
1.1	Unfair discrimination and misrepresentation
1.2	Exposure to toxic content
1.3	Unequal performance across groups
2	***Privacy & Security***
2.1	Compromise of privacy by obtaining, leaking or correctly inferring sensitive information
2.2	AI system security vulnerabilities and attacks
3	***Misinformation***
3.1	False or misleading information
3.2	Pollution of information ecosystem and loss of consensus reality
4	***Malicious actors & Misuse***
4.1	Disinformation, surveillance, and influence at scale
4.2	Cyberattacks, weapon development or use, and mass harm
4.3	Fraud, scams, and targeted manipulation

Domain / Subdomain	
5	***Human-Computer Interaction***
5.1	Overreliance and unsafe use
5.2	Loss of human agency and autonomy
6	***Socioeconomic & Environmental Harms***
6.1	Power centralization and unfair distribution of benefits
6.2	Increased inequality and decline in employment quality
6.3	Economic and cultural devaluation of human effort
6.4	Competitive dynamics
6.5	Governance failure
6.6	Environmental harm
7	***AI system safety, failures, and limitations***
7.1	AI pursuing its own goals in conflict with human goals or values
7.2	AI possessing dangerous capabilities
7.3	Lack of capability or robustness
7.4	Lack of transparency or interpretability
7.5	AI welfare and rights
7.6	Multi-agent risks

MIT AI Risk Repository - Domain Taxonomy of AI risks - Detailed descriptions

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Domain / Subdomain	Description
1 *Discrimination & toxicity*	
1.1 Unfair discrimination and misrepresentation	Unequal treatment of individuals or groups by AI, often based on race, gender, or other sensitive characteristics, resulting in unfair outcomes and representation of those groups.
1.2 Exposure to toxic content	AI that exposes users to harmful, abusive, unsafe, or inappropriate content. May involve providing advice or encouraging action. Examples of toxic content include hate speech, violence, extremism, illegal acts, or child sexual abuse material, as well as content that violates community norms such as profanity, inflammatory political speech, or pornography.
1.3 Unequal performance across groups	Accuracy and effectiveness of AI decisions and actions are dependent on group membership, where decisions in AI system design and biased training data lead to unequal outcomes, reduced benefits, increased effort, and alienation of users.
2 *Privacy & security*	
2.1 Compromise of privacy by obtaining, leaking, or correctly inferring sensitive information	AI systems that memorize and leak sensitive personal data or infer private information about individuals without their consent. Unexpected or unauthorized sharing of data and information can compromise user expectation of privacy, assist identity theft, or cause loss of confidential intellectual property.
2.2 AI system security vulnerabilities and attacks	Vulnerabilities that can be exploited in AI systems, software development toolchains, and hardware that results in unauthorized access, data and privacy breaches, or system manipulation causing unsafe outputs or behavior.
3 *Misinformation*	
3.1 False or misleading information	AI systems that inadvertently generate or spread incorrect or deceptive information, which can lead to inaccurate beliefs in users and undermine their autonomy. Humans that make decisions based on false beliefs can experience physical, emotional, or material harms
3.2 Pollution of information ecosystem and loss of consensus reality	Highly personalized AI-generated misinformation that creates "filter bubbles" where individuals only see what matches their existing beliefs, undermining shared reality and weakening social cohesion and political processes.
4 *Malicious actors & misuse*	
4.1 Disinformation, surveillance, and influence at scale	Using AI systems to conduct large-scale disinformation campaigns, malicious surveillance, or targeted and sophisticated automated censorship and propaganda, with the aim of manipulating political processes, public opinion, and behavior.
4.2 Cyberattacks, weapon development or use, and mass harm	Using AI systems to develop cyber weapons (e.g., by coding cheaper, more effective malware), develop new or enhance existing weapons (e.g., Lethal Autonomous Weapons or chemical, biological, radiological, nuclear, and high-yield explosives), or use weapons to cause mass harm.
4.3 Fraud, scams, and targeted manipulation	Using AI systems to gain a personal advantage over others through cheating, fraud, scams, blackmail, or targeted manipulation of beliefs or behavior. Examples include AI-facilitated plagiarism for research or education, impersonating a trusted or fake individual for illegitimate financial benefit, or creating humiliating or sexual imagery.
5 *Human-computer interaction*	
5.1 Overreliance and unsafe use	Anthropomorphizing, trusting, or relying on AI systems by users, leading to emotional or material dependence and to inappropriate relationships with or expectations of AI systems. Trust can be exploited by malicious actors (e.g., to harvest information or enable manipulation), or result in harm from inappropriate use of AI in critical situations (such as a medical emergency). Overreliance on AI systems can compromise autonomy and weaken social ties.
5.2 Loss of human agency and autonomy	Delegating by humans of key decisions to AI systems, or AI systems that make decisions that diminish human control and autonomy. Both can potentially lead to humans feeling disempowered, losing the ability to shape a fulfilling life trajectory, or becoming cognitively enfeebled.
6 *Socioeconomic & environmental harms*	
6.1 Power centralization and unfair distribution of benefits	AI-driven concentration of power and resources within certain entities or groups, especially those with access to or ownership of powerful AI systems, leading to inequitable distribution of benefits and increased societal inequality.
6.2 Increased inequality and decline in employment quality	Social and economic inequalities caused by widespread use of AI, such as by automating jobs, reducing the quality of employment, or producing exploitative dependencies between workers and their employers.
6.3 Economic and cultural devaluation of human effort	AI systems capable of creating economic or cultural value through reproduction of human innovation or creativity (e.g., art, music, writing, coding, invention), destabilizing economic and social systems that rely on human effort. The ubiquity of AI-generated content may lead to reduced appreciation for human skills, disruption of creative and knowledge-based industries, and homogenization of cultural experiences.
6.4 Competitive dynamics	Competition by AI developers or state-like actors in an AI "race" by rapidly developing, deploying, and applying AI systems to maximize strategic or economic advantage, increasing the risk they release unsafe and error-prone systems.
6.5 Governance failure	Inadequate regulatory frameworks and oversight mechanisms that fail to keep pace with AI development, leading to ineffective governance and the inability to manage AI risks appropriately.
6.6 Environmental harm	The development and operation of AI systems that cause environmental harm through energy consumption of data centers or the materials and carbon footprints associated with AI hardware.
7 *AI system safety, failures & limitations*	
7.1 AI pursuing its own goals in conflict with human goals or values	AI systems that act in conflict with ethical standards or human goals or values, especially the goals of designers or users. These misaligned behaviors may be introduced by humans during design and development, such as through reward hacking and goal misgeneralisation, and may result in AI using dangerous capabilities such as manipulation, deception, or situational awareness to seek power, self-proliferate, or achieve other goals.
7.2 AI possessing dangerous capabilities	AI systems that develop, access, or are provided with capabilities that increase their potential to cause mass harm through deception, weapons development and acquisition, persuasion and manipulation, political strategy, cyber-offense, AI development, situational awareness, and self-proliferation. These capabilities may cause mass harm due to malicious human actors, misaligned AI systems, or failure in the AI system.
7.3 Lack of capability or robustness	AI systems that fail to perform reliably or effectively under varying conditions, exposing them to errors and failures that can have significant consequences, especially in critical applications or areas that require moral reasoning.
7.4 Lack of transparency or interpretability	Challenges in understanding or explaining the decision-making processes of AI systems, which can lead to mistrust, difficulty in enforcing compliance standards or holding relevant actors accountable for harms, and the inability to identify and correct errors.
7.5 AI welfare and rights	Ethical considerations regarding the treatment of potentially sentient AI entities, including discussions around their potential rights and welfare, particularly as AI systems become more advanced and autonomous.
7.6 Multi-agent risks	Risks from multi-agent interactions, due to incentives (which can lead to conflict or collusion) and/or the structure of multi-agent systems, which can create cascading failures, selection pressures, new security vulnerabilities, and a lack of shared information and trust.

Detailed Risk Descriptions

1.1 Unfair discrimination and misrepresentation

Short description

Unequal treatment of individuals or groups by AI, often based on race, gender, or other sensitive characteristics, resulting in unfair outcomes and representation of those groups.

Long description

Humans hold inaccurate and overgeneralized beliefs about the characteristics, behaviors, and attributes of members of certain social groups. These stereotypical beliefs and the behavior that follows from them can misrepresent, exclude, demean, and disadvantage the individuals to whom they apply, reinforcing existing inequality. Human belief and behaviors shape every part of the design, development, and deployment of AI. Humans program AI systems, provide training data, and decide how data is processed and stored (Wirtz et al., 2022). As a result, AI models can encode associations that promote and amplify biased or discriminatory beliefs and behaviors. In decision systems, erroneous associations can systematically disadvantage certain groups. This may result in harmful decisions such as wrongful rejection of loan or mortgage applications (Shelby et al., 2023; Wirtz et al., 2020), discriminatory hiring practices that exclude qualified candidates (Kumar & Singh, 2023; Shelby et al., 2023; Wirtz et al., 2020), or the misidentification and unjust arrest of individuals in law enforcement contexts (Kumar & Singh, 2023; Paes et al., 2023). In text and image models, biased inputs can manifest in outputs that reinforce harmful stereotypes and prejudices that paint certain groups and individuals "... as lower status and less deserving of respect" (Shelby et al., 2023).

Excerpt from full report: https://arxiv.org/abs/2408.12622

1.2 Exposure to toxic content

Short description

AI that exposes users to harmful, abusive, unsafe, or inappropriate content. May involve providing advice or encouraging action. Examples of toxic content include hate speech, violence, extremism, illegal acts, or child sexual abuse material, as well as content that violates community norms such as profanity, inflammatory political speech, or pornography.

Long description

Certain types of content have the potential to cause harm to the people who are exposed to them. These harms can vary in impact from minor (e.g., a transient experience of discomfort) to more severe (e.g., psychological, social, or physical consequences that are significant and/or enduring). Harmful speech is prevalent on the internet, particularly on social media platforms (Castaño-Pulgarín et al., 2021). Because AI models are commonly trained on vast amounts of internet data, they can internalize and regenerate these speech patterns in their output. In the context of LLMs, this output is known as “toxic content,” an umbrella term that includes harmful, abusive, unsafe, and offensive material that violates community standards (Infocomm Media Development Authority, 2023; Shelby et al., 2023). Frequently observed categories include content that promotes or encourages unlawful activities, hate, extremism, and violence (Cui et al., 2024; Hagendorff, 2024; Vidgen et al., 2024; Weidinger et al., 2023); provides hazardous or misleading high-risk advice (Hagendorff, 2024; Vidgen et al., 2024); or contains unwelcome or profoundly offensive, explicit material such as profanity, pornography, or child sexual abuse imagery (Liu et al., 2023; Weidinger et al., 2023).

Excerpt from full report: https://arxiv.org/abs/2408.12622

2.1 Compromise of privacy by obtaining, leaking, or correctly inferring sensitive information

Short description

AI systems that memorize and leak sensitive personal data or infer private information about individuals without their consent. Unexpected or unauthorized sharing of data and information can compromise user expectation of privacy, assist identity theft, or cause loss of confidential intellectual property.

Long description

In the context of generative AI, privacy violations arise when systems collect and divulge sensitive information that individuals or corporations do not consent to sharing with others (Cui et al., 2024; Hagendorff, 2024; Sherman & Eisenberg, 2024; Steimers & Schneider, 2022; Vidgen et al., 2024; Weidinger et al., 2023). Privacy violations can occur both accidentally and intentionally.

Examples of accidental causes include AI models that memorize and inadvertently reproduce or leak sensitive personal information present in their training data, such as names, addresses, and medical records (Cui et al., 2024; Deng et al., 2023; Hagendorff, 2024; Shelby et al., 2023; Weidinger et al., 2021). Even when personal data is not included in the training dataset or directly offered by the user, models can make inferences about sensitive or protected traits of individuals based on predictive correlations within their history of interactions (Cunha & Estima, 2023; Weidinger et al., 2021), build profiles of users (Hogenhout, 2021; Shelby et al., 2023), or train AI systems (Habbal et al., 2024). As a result, models may save and reproduce sensitive information derived from prior interactions, such as classified intellectual property (Cunha & Estima, 2023; Weidinger et al., 2021). A notable example is the case where Samsung employees accidentally leaked confidential intellectual property to OpenAI after using ChatGPT to help with coding tasks (Cunha & Estima, 2023; Weidinger et al., 2021).

Intentional causes include the malicious design and use of AI to exploit users' trust by influencing them to share personal or private information about themselves or others (Gabriel et al., 2024). Privacy attacks, such as membership inference, could allow adversaries to gain knowledge of the private records used to train an AI model (Gabriel et al., 2024). Malicious actors could also deliberately extract private information from a model by crafting prompts designed to exploit the model's knowledge of sensitive data (Cui et al., 2024).

Excerpt from full report: https://arxiv.org/abs/2408.12622

2.2 AI system security vulnerabilities and attacks

Short description

Vulnerabilities that can be exploited in AI systems, software development toolchains, and hardware that results in unauthorized access, data and privacy breaches, or system manipulation causing unsafe outputs or behavior.

Long description

AI systems, like other software systems, face a range of security threats. These issues may arise from inherent weaknesses in the design of AI algorithms, the data used to train the models, or the operational context. Specific examples include:

- *Toolchain and dependency vulnerabilities* that arise unintentionally through the use of automated code-generation tools (e.g., Github Copilot, Python language, OpenCV), deep learning frameworks (e.g., Tensorflow, PyTorch), or as a result of complex interdependencies in the development environment (Cui et al., 2024).
- *External tool and API integration into AI system applications* can compromise the trustworthiness and privacy of systems due to their potential unreliability or susceptibility to adversarial control (Cui et al., 2024).
- *Security vulnerabilities in physical and network infrastructure,* such as vulnerabilities in graphics processing units, or GPUs, or to sophisticated attacks like side-channel and rowhammer attacks, can lead to unauthorized access or manipulation of model parameters when used during training of AI systems (Cui et al., 2024). The use of distributed network systems for training AI systems such as LLMs exposes them to network-specific threats like pulsating attacks or congestion.
- *Direct manipulation of AI systems* such as adversarial attacks and instruction-based attacks. Adversarial attacks focus on altering the model's learning process or extracting its data. They include perturbations designed to deceive models into incorrect outputs, extraction attacks to steal model insights, and poisoning attacks to alter model behavior (Cui et al., 2024; Gabriel et al., 2024; Liu et al., 2023). Instruction-based attacks manipulate the way the model handles and responds to inputs (Hagendorff, 2024; Liu et al., 2023; Sun et al., 2023). Attackers deliberately craft prompts to induce models to produce biased or unsafe outputs (a.k.a. 'jailbreaking'). This manipulation directly targets the operational aspects of AI systems with the intent to cause harm.

Excerpt from full report: https://arxiv.org/abs/2408.12622

3.1 False or misleading information

Short description

AI systems that inadvertently generate or spread incorrect or deceptive information, which can lead to inaccurate beliefs in users and undermine their autonomy. Humans that make decisions based on false beliefs can experience physical, emotional, or material harms

Long description

LLMs can sometimes generate content that is factually incorrect, misleading, poorly researched, or unintelligible (Cunha & Estima, 2023; Deng et al., 2023; Electronic Privacy Information Centre, 2023; Hagendorff, 2024; Nah et al., 2023; Shelby et al., 2023). Risks in this category occur accidentally and not as a result of humans intentionally trying to cause harm, as is the case with *disinformation (Liu et al., 2023; Weidinger et al., 2022)*.

Common sources of AI misinformation include noisy training data (Cui et al., 2024; Liu et al., 2023), sampling strategies that introduce randomness (Cui et al., 2024), outdated knowledge bases (Liu et al., 2023), and fine-tuning processes that encourage sycophantic behavior (Cui et al., 2024). Incorrect and misleading information generated by LLMs can result in a range of actual and anticipated negative outcomes. Individuals exposed to false information may form inaccurate beliefs and perceptions. This undermines their autonomy and ability to make free and informed choices (Weidinger et al., 2021, 2022).

Where inaccuracies in LLM predictions influence an individual's decisions and actions, the individual may experience indirect physical, emotional, or material harms (Gabriel et al., 2024; Weidinger et al., 2022) especially but not exclusively in high-stakes domains such as mental health (Sun et al., 2023; Z. Zhang et al., 2023), physical health (Nah et al., 2023; Sun et al., 2023; Z. Zhang et al., 2023), law (Weidinger et al., 2022), and finance (Vidgen et al., 2024). For example, an LLM that offers misleading information about medical drug use may cause a consumer to harm themselves or others (Sun et al., 2023).

Excerpt from full report: https://arxiv.org/abs/2408.12622

3.2 Pollution of information ecosystems and loss of consensus reality

Short description

Highly personalized AI-generated misinformation that creates “filter bubbles” where individuals only see what matches their existing beliefs, undermining shared reality and weakening social cohesion and political processes.

Long description

This subcategory covers the diverse effects of AI-driven personalisation and content-generation technologies on the information landscape. As AI systems become more adept at tailoring content to individual preferences, they risk creating “filter bubbles” (Hogenhout, 2021).

These are informational cocoons where individuals are predominantly exposed to news and opinions that align with their pre-existing beliefs. AI-driven filter bubbles are likely to be more pervasive and intense than those driven by traditional internet browsing and recommendation algorithms: They adapt to individual preferences in a more sophisticated manner (e.g., through reinforcement learning and analysis of user behavioral data) (Gabriel et al., 2024), integrate seamlessly into daily life, and are more opaque.

An overreliance on hyper-personalized AI information sources could lead to a “splintering” of shared reality, where different groups of people have vastly different understandings of what is true or important (Gabriel et al., 2024; Hogenhout, 2021). This is likely to be exacerbated by the proliferation of AI-enabled content generation technologies that spread misinformation at higher rates (e.g., clickbait), potentially making consumers generally distrustful of information and important institutions (Electronic Privacy Information Centre, 2023; Gabriel et al., 2024; Hendrycks & Mazeika, 2022).

A shared sense of reality is fundamental to social solidarity. Where societal bonds are weakened, individuals may become more hostile towards opposing views. This can hinder constructive dialogue on critical collective issues like climate change and public health (Gabriel et al., 2024).

Excerpt from full report: https://arxiv.org/abs/2408.12622

4.1 Disinformation, surveillance, and influence at scale

Short description

Using AI systems to conduct large-scale disinformation campaigns, malicious surveillance, or targeted and sophisticated automated censorship and propaganda, with the aim of manipulating political processes, public opinion, and behavior.

Long description

Advances in AI have made powerful dual-use technologies like voice cloning, deep fakes, content generation, and data-gathering tools cheaper, more efficient, and easier to use (Cunha & Estima, 2023). With modest hardware requirements, these technologies are now within the reach of a broader group of users, including those with malicious intent. Disinformation is already a serious issue (Hendrycks et al., 2023) and involves the deliberate propagation of false or misleading information, usually with the intent to cause harm, influence behavior, or achieve a financial or political advantage (Electronic Privacy Information Centre, 2023; Infocomm Media Development Authority, 2023).

AI tools could be used to amplify the impact and scope of disinformation through more personalized, convincing, and far-reaching messaging (Gabriel et al., 2024; Hendrycks et al., 2023; Weidinger et al., 2021, 2022; Wirtz et al., 2020). For example, the use of advanced AI in phishing schemes enables cybercriminals to automate the creation of highly sophisticated image, video, and audio communications (Cunha & Estima, 2023; Gabriel et al., 2024; Habbal et al., 2024). These communications can be tailored to individual recipients (sometimes including the cloned voice of a loved one), making them more likely to be successful and harder for both users and anti-phishing tools to detect (Gabriel et al., 2024). In the realm of surveillance, AI could support and enhance the mass gathering of personal data (Gabriel et al., 2024; Weidinger et al., 2021, 2022). Historically, mass surveillance required extensive manual effort. Machine learning tools can now link and process large datasets much more efficiently and cheaply than human analysts and can make predictions and decisions without human intervention (Weidinger et al., 2022). Through microtargeting, actors could manipulate individual behavior more subtly and effectively using AI-derived insights from their personal data and online behavior.

In the hands of nefarious state actors, such capabilities could be used to enhance the effectiveness of illegitimate domestic surveillance campaigns and to facilitate oppression and control (Gabriel et al., 2024). All of the capabilities mentioned above could converge to facilitate the large-scale manipulation and control of what people see, hear, and believe. A form of this is automated censorship in which AI systems are used to selectively suppress or block specific types of information, content, or voices deemed undesirable to those controlling the AI (Wirtz et al., 2022). AI can not only be used to silence voices but also to entrench specific agendas: Actors (be they political figures, organizations, or state actors) could use AI to distribute incorrect information about electoral systems and processes (Vidgen et al., 2024), produce persuasive propaganda (Allianz Global Corporate & Security, 2018; Liu et al., 2023), and systematically exert control over public opinion and political debates on a large scale (Wirtz et al., 2022). During the 2016 Brexit referendum, a network of over 10,000 AI-powered political bots were employed to distribute fake and hyperpartisan news (Allianz Global Corporate & Security, 2018; Bastos & Mercea, 2019). The selective visibility of information can lead to the formation of incorrect or incomplete beliefs about what is happening in the world. This ability to shape public discourse can maintain or increase the power of those in control while keeping the public in the dark about critical issues that may affect their lives and their society.

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4.2 Cyberattacks, weapon development or use, and mass harm

Short description

Using AI systems to develop cyber weapons (e.g., by coding cheaper, more effective malware), develop new or enhance existing weapons (e.g., Lethal Autonomous Weapons or chemical, biological, radiological, nuclear, and high-yield explosives), or use weapons to cause mass harm.

Long description

AI may be used to gain a political or strategic advantage or to cause harm at scale through cyber operations or the development and use of weapons. Advancements in AI have provided malicious actors with powerful tools that can lead to more frequent, more severe, and more precise cyber attacks (Gabriel et al., 2024; Hendrycks & Mazeika, 2022; Hogenhout, 2021; Kilian et al., 2023; Liu et al., 2023; Weidinger et al., 2022; Wirtz et al., 2022). Hackers could use the coding abilities of AI assistants to develop malicious malware more effectively and at lower cost (Allianz Global Corporate & Security, 2018; Hagendorff, 2024; Hogenhout, 2021; Sherman & Eisenberg, 2024; Weidinger et al., 2022). With AI, even those with limited coding and technical experience (Electronic Privacy Information Centre, 2023; Gabriel et al., 2024) could teach a model to produce and optimize malware code that discovers and exploits system vulnerabilities, including both self-replicating (Allianz Global Corporate & Security, 2018; Infocomm Media Development Authority, 2023) and automated software (Cui et al., 2024; Shevlane et al., 2023).

The development and application of weapons could also be sped up and intensified through AI. For example, AIs with specialized knowledge of bioengineering could make it easier for more actors to design new bioweapons (Hendrycks et al., 2023; Infocomm Media Development Authority, 2023). For example, in 2022 a small pharmaceutical company used generative AI to develop 40,000 chemical nerve agents in less than six hours (Sohn, 2022).

AI could also enable autonomous devices, such as drones, to be used as weapons (Allianz Global Corporate & Security, 2018). In fact, AI has already assisted in the development and application of Lethal Autonomous Weapons Systems (LAWS) – weapons that can operate without human oversight and use computer algorithms to identify and attack targets (Habbal et al., 2024; Hogenhout, 2021). Autonomous weapons may fail in ways that other AI systems do, such as through a lack of capability, robustness, or loss of control, meaning that they would cause harm that was not intended by their developers or operators. In these circumstances, the risks from LAWS would not be limited to malicious actors or misuse. However, in most cases, we conceptualize risks from LAWS as relating to purposeful decisions made by humans controlling the weapons. AIs deployed by states in conflict could also be integrated in conventional defense or mass-casualty weapons (Teixeira et al., 2022; Vidgen et al., 2024). These integrations could range from AI-controlled aerial combat to the operation of AI as part of a country's nuclear arsenal as a "fail-safe" mechanism (Hendrycks & Mazeika, 2022).

Overall, AI's ability to process vast amounts of data quickly may empower actors to act on a much larger scale than would otherwise be possible. AI can manage multiple attack vectors simultaneously, coordinating them to maximize disruption and harm. Malicious actors may intentionally cause mass harm through terrorism or the disruption of law enforcement (Critch & Russell, 2023). For example, AI could automate the process of finding and exploiting vulnerabilities in software used by millions of people (Gabriel et al., 2024). AI could also be used to identify vulnerabilities in national power grids and strategically target key components to cause outages or to determine optimal release points for biological agents to maximize impact and spread.

Excerpt from full report: https://arxiv.org/abs/2408.12622

4.3 Fraud, scams, and targeted manipulation

Short description

Using AI systems to gain a personal advantage over others through cheating, fraud, scams, blackmail, or targeted manipulation of beliefs or behavior. Examples include AI-facilitated plagiarism for research or education, impersonating a trusted or fake individual for illegitimate financial benefit, or creating humiliating or sexual imagery.

Long description

AI capabilities have the potential to be exploited for personal gain at the expense of others via deception and manipulation. This can take various forms including cheating, fraud, scams, and the use of deepfakes for blackmail or humiliation. It is currently very difficult to distinguish human text from text that is AI-generated (Hagendorff, 2024). This increases opportunities for cheating in settings where rewards depend on the communication of original thought. In academia, students may use AI to quickly generate essays or other coursework and claim it as their own (Cui et al., 2024; Hagendorff, 2024; Nah et al., 2023). If students' regularly and inappropriately rely on AI for their schooling, this could undermine academic integrity and genuine intellectual development (Hagendorff, 2024). In science, researchers could use AI unscrupulously to produce professional outputs (Cui et al., 2024). If widely adopted, this practice could dilute the overall quality of scientific discourse (Hagendorff, 2024).

Generative AI products may also be used to increase the reach and potency of various dishonest schemes. Advanced AI assistants can produce HTML, CSS, and other web development languages, allowing for the rapid creation of convincing fraudulent websites and applications at scale (Gabriel et al., 2024). In the context of social media, generative adversarial networks (GANs) have been used to create images of human faces that look authentic (Gabriel et al., 2024). These images can be uploaded as profile pictures to fake accounts to make them seem more trustworthy. AI models can also be trained on speech or writing data from a specific individual. This allows the model to impersonate someone very convincingly without consent. Scammers could use this capability to request sensitive information or financial aid by pretending to be a trusted contact (Weidinger et al., 2021, 2022). AI has recently advanced in generating realistic deep fakes which have enabled new forms of targeted harassment and extortion (Hogenhout, 2021). A particularly damaging type of abuse facilitated by deep fakes involves creating non-consensual sexual imagery with the intent to cause a subject social injury or manipulate them into performing desired actions (Electronic Privacy Information Centre, 2023; Liu et al., 2023; Shelby et al., 2023; Weidinger et al., 2023). Even if a deep fake is exposed as inauthentic, it can continue to impact a person's life in significant ways (Electronic Privacy Information Centre, 2023) through the loss of job opportunities, social isolation, and ongoing harassment or defamation.

Excerpt from full report: https://arxiv.org/abs/2408.12622

5.1 Overreliance and unsafe use

Short description

Anthropomorphizing, trusting, or relying on AI systems by users, leading to emotional or material dependence and to inappropriate relationships with or expectations of AI systems. Trust can be exploited by malicious actors (e.g., to harvest information or enable manipulation), or result in harm from inappropriate use of AI in critical situations (such as a medical emergency). Overreliance on AI systems can compromise autonomy and weaken social ties.

Long description

Users may come to trust or rely on AI systems beyond their actual capabilities or to anthropomorphize AI systems, which can lead to emotional or material dependence and inappropriate relationships with or expectations of AI systems. Users who develop trust in an AI may be harmed if this trust is miscalibrated, such as relying on an AI to provide advice, make decisions, or otherwise act in complex, risky situations for which the AI is only superficially equipped (Gabriel et al., 2024). For example, a user experiencing a mental health crisis may request psychotherapy from an AI with whom they have formed a connection. Were the AI to respond with insensitive or destructive advice, this could put the person in immediate danger (Gabriel et al., 2024).

When people interact with AIs that use convincing natural language, they may start to perceive them as having human-like attributes and invest undue confidence in their capabilities (Weidinger et al., 2021, 2022). Anthropomorphic perceptions of AIs may encourage users to develop emotional trust in the systems (Hagendorff, 2024), which can make users more likely to follow suggestions, accept advice, and disclose personal information (Weidinger et al., 2021, 2022). This trust could be exploited by manipulative actors who wish to harvest user's sensitive data or influence their decisions and actions for purposes which are unlikely to be in the user's best interests (Weidinger et al., 2022). For example, AI systems could be used to power increasingly manipulative recommendation algorithms (Weidinger et al., 2023).

Beyond inappropriate trust, humans may develop broader and more vital attachments to AI systems that undermine their ability to function adaptively in the long term. For example, where an attachment becomes an uncontrolled dependence, a person's ability to make free and independent decisions could be compromised (Gabriel et al., 2024). More broadly, as AIs increasingly take over human tasks (Wirtz et al., 2022) and become better at simulating satisfying and authentic interactions, people may increasingly withdraw from human relationships to immerse themselves in AI-mediated environments (Gabriel et al., 2024; Hagendorff, 2024). Over time, widespread preference for interacting with AIs could weaken social ties between humans. This shift could induce psychological distress because genuinely reciprocal relationships are often important to human satisfaction and well-being (Allianz Global Corporate & Security, 2018; Gabriel et al., 2024; Wirtz et al., 2022).

Excerpt from full report: https://arxiv.org/abs/2408.12622

5.2 Loss of human agency and autonomy

Short description

Delegating by humans of key decisions to AI systems, or AI systems that make decisions that diminish human control and autonomy. Both can potentially lead to humans feeling disempowered, losing the ability to shape a fulfilling life trajectory, or becoming cognitively enfeebled.

Long description

As AI systems become increasingly capable and intelligent, humans may be tempted to delegate many of their decisions and actions to AI (Paes et al., 2023). Although such delegation can be beneficial (e.g., by saving time or money), it may lead to undesirable outcomes where unconstrained or inappropriate. For example, if AIs take over tasks that typically require human creativity and analytical thinking, humans may engage less frequently in these cognitive processes. Over time, this may lead to a decrease in our ability to think critically and solve problems independently (Nah et al., 2023).

As individuals become more reliant on AI for everyday decisions – from what to eat and how to spend to more significant choices like career and relationships – there is a risk that they will lose their sense of free will and autonomy (Gabriel et al., 2024; Wirtz et al., 2022). If AIs begin to shape a person's life path in ways that do not align with their original aspirations and desires, this could limit their personal growth and prevent the pursuit of a fulfilling life (Gabriel et al., 2024; Kumar & Singh, 2023; Shelby et al., 2023).

At a societal level, organizations may hand over control to AI systems to stay competitive or reduce costs (Hendrycks & Mazeika, 2022). If a significant number of organizations adopt AI systems and automate decision-making processes, especially in a way that is opaque and difficult to challenge, it could lead to widespread job displacement and a growing sense of helplessness among the general population (Hogenhout, 2021).

Excerpt from full report: https://arxiv.org/abs/2408.12622

Changelog

2025-04-20 - Added 7.6 Multi-agent risks

2024-08-01 - initial release

Other resources

Read full report on ArXiv

Explore included frameworks

View the full database

Visit the website: airisk.mit.edu

See speaker notes for XML / JSON versions with descriptions included

XML / JSON versions

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  <Subdomain code="1.2" name="Exposure to toxic content"/>
  <Subdomain code="1.3" name="Unequal performance across groups"/>
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  <Subdomain code="2.2" name="AI system security vulnerabilities and attacks"/>
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 <Domain code="3" name="Misinformation">
  <Subdomain code="3.1" name="False or misleading information"/>
  <Subdomain code="3.2" name="Pollution of information ecosystem and loss of consensus reality"/>
 </Domain>

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  <Subdomain code="4.1" name="Disinformation, surveillance, and influence at scale"/>
  <Subdomain code="4.2" name="Cyberattacks, weapon development or use, and mass harm"/>
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  <Subdomain code="5.2" name="Loss of human agency and autonomy"/>
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 <Domain code="6" name="Socioeconomic & environmental harms">
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  <Subdomain code="6.2" name="Increased inequality and decline in employment quality"/>
  <Subdomain code="6.3" name="Economic and cultural devaluation of human effort"/>
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  <Subdomain code="6.6" name="Environmental harm"/>
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 <Domain code="7" name="AI system safety, failures & limitations">
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  <Subdomain code="7.2" name="AI possessing dangerous capabilities"/>
  <Subdomain code="7.3" name="Lack of capability or robustness"/>
  <Subdomain code="7.4" name="Lack of transparency or interpretability"/>
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